ME3 | Detailed system architecture

Standalone Raspberry Pi 5 runtime with Wi-Fi | Voice, text, shared context and guarded task execution

CLOUD SERVICES — model inference and authorized external access

Cloud speech recognition

Groq · whisper-large-v3-turbo
WAV audio over HTTPS → transcript
Recognition model runs outside the Pi

Cloud reasoning — distinct model roles

Main Onit task reasoning: gpt-6.1-sol
Host floor advisory: qwen/qwen3.8-27b, tool-free
Separate roles; Qwen cannot authorize actions
Admitted / counted provider requests

Configured internet services

Web research · Google Calendar · Gmail
WhatsApp and supported media routes
Access only within configured scope

RASPBERRY PI 5 — local services, conversation ownership and execution control

ME3 GUI: microphone start/stop · text input · conversation / Copy conversation · status notices · Artifacts preview

1 Capture and audio processing

Pi microphone → Pipecat pipeline
Local Silero VAD / SmartTurn
WebRTC AEC and turn / pause handling
Listening remains available during output

captured audio

2 Student-built ASR MCP service

FastMCP / stdio · transcribe(...)
16 kHz mono PCM16, Base64 input
Validate → WAV / Groq → recognized text
Request IDs, deadlines and cleanup

3 Shared conversation and task context

One conversation owner for voice AND typed input
Bounded history · goal ID / revision · clarification
Tool evidence · pending review · retained replies
Current artifact / preview / service references
Persistent artifacts are read through tools

context / response history

4 Onit — sole task-planning loop

Interprets the goal using the selected language model
Chooses supplied tools and prepares responses
Provider broker checks admission / counts requests
No model-generated expansion of host permissions

verified reply delivery

5 Guarded execution gateway

Validate permitted action and current identity
Exact recipient / content / action review
Separate confirmation when required
Verify results before claiming completion
Uncertain outcome → readback, not replay

tools / verified results

6 Pi-hosted MCP tools and adapters

Research · utilities · local media
Calendars · documents · communication
Artifacts / code workspace · bounded jobs
Actual tool schemas and scoped access
Results return through the gateway

files / service state

7 Validated speech delivery

Complete reply or validated earlier speech prefix
Spoken-text preparation and coherent phrase groups
Bounded next-phrase synthesis during playback
Current-generation ownership at delivery boundaries
A progress cue is not proof of task completion

text + language

8 Student-built TTS MCP service

FastMCP / stdio · synthesize(...)
Resident local Piper CPU worker
English: en_US-ljspeech-medium
Female Spanish: es_ES-sharvard-medium
Returns 24 kHz mono PCM16 audio

PCM for playback

9 Speech / music mixer and speaker

Generation-owned playback buffers
Supported local / selected media playback
Speaker output and AEC reference
Pause / resume / clear obsolete output

Dotted return: mixer output reference → AEC

Artifacts, previews and owned jobs

Saved files / apps and current file versions
Preview opens inside the ME3 GUI
Bounded background server + health status
Close preview ≠ stop its server

Durable state and verification evidence

Action journals / once-only intent records
Counters, status and owned-job evidence
No automatic resend of uncertain effects

Duplex control — host-owned, not a second planner

Speech activity → contextual floor interpretation
Acknowledgment: suspend / resume the SAME answer
Correction: invalidate stale generation; replace reply
Generation + request IDs fence obsolete results
Tool-free Qwen advisory supports floor decisions
Host validates; advisory cannot authorize actions

Separate tool-free floor advisory

"Student-built" means the MCP service and integration were implemented for the assignment; the speech models were reused.

Cloud calls require Wi-Fi and private provider/account setup. The normal Pi runtime does not require the laptop or Codex desktop.

Floor advisory: qwen/qwen3.8-27b, tool-free. Startup prewarming remains inactive. External actions remain guarded and independently verified.

READING THE CONNECTIONS

- Audio, text, model and tool data; two arrowheads indicate exchange
- Duplex control: activity, ownership, pause / resume and cancellation
- State / verification records or audio echo reference

Aligned with the two-page summary and expanded script · Pi source 03d7a144… · Architecture diagram, not a claim that every case passed.

ME3 | Architecture explained

How to read page 1: roles, boundaries, conversation and guarded execution

1. Read the Pi / cloud boundary

The upper band contains cloud recognition, cloud reasoning and configured internet services. The large enclosure contains Pi-local processes, conversation state and execution control.

The normal runtime uses the Pi display, microphone, speaker and Wi-Fi. No laptop or Codex desktop runtime process is required. Private provider/account setup is still necessary.

2. Follow the speech and text paths

Voice: microphone / VAD → student ASR MCP → shared context → Onit → validated delivery → student TTS MCP → mixer / speaker. Tool-backed requests branch through the guarded gateway and return verified results.

Typed text is also a valid input. It joins the same conversation context and can supply a whole request, clarify unclear speech or correct a misrecognized name. Enter/Send submits; Shift+Enter inserts a newline. It bypasses ASR for that input, not permissions or model/tool connectivity.

Recognized speech and replies appear in the GUI. Copy conversation copies that displayed exchange; it does not export raw audio or every internal event. Displayed reply text does not prove that all words were audible.

3. Interpret the connections

Solid teal: audio, text, model or tool data. Two arrowheads indicate an exchange.

Dashed orange: duplex control and the separate tool-free floor advisory. Dotted gray: verification/state records or the mixer reference used by AEC, according to the node label.

Crossing lines without a junction dot do not imply a shared connection. The figure shows component relationships, not a separate process or network link for every arrow.

4. Distinguish reused models from student services

Student-built means the Python MCP service and integration were implemented for the assignment; Whisper, Piper and Onit were reused. FastMCP/stdio exposes structured speech tools locally.

ASR accepts Base64-encoded 16 kHz mono PCM16 and calls Groq Whisper large-v3-turbo. Piper runs locally in a resident worker and returns 24 kHz mono PCM16, with the selected English and female Spanish voices.

Owner trials demonstrated Spanish drafting/translation and contextual Spanish replies without a language button. The main reasoner supplies language-tagged answer segments; validated en/es tags select the configured Piper voice. English and Spanish use different voices, not one bilingual voice.

Recognition remains English-configured; Spanish input can be captured as English or need a typed clarification. Translation quality and Spanish recognition/pronunciation were not formally benchmarked. No additional spoken language is qualified.

5. Keep reasoning roles and safeguards separate

Onit is the sole task-planning loop using the selected gpt-6.1-sol main reasoner. Qwen qwen3.8-27b supplies a separate, tool-free cloud floor advisory. The Pi host validates it; it cannot authorize actions or become a fallback task planner.

Acknowledgments can suspend and resume the same answer. Corrections can invalidate obsolete output. Generation/request IDs support ownership checks and rejection of stale results.

The gateway validates scope, exact review and separate confirmation where required. Results need verification before a success claim. Uncertain or completed effects are not automatically replayed, and interruption does not undo a committed action.

6. Understand state, previews and evidence

Context contains bounded working history and task references, not unlimited memory. Saved artifacts are persistent files read through tools. In-GUI previews and owned background services have separate state: closing a preview does not stop its server.

Journals retain once-only outcomes and accounting. A progress cue is not task completion. Startup prewarming remains inactive.

Architecture mechanisms and scoped live demonstrations do not establish universal success or measured physical stop latency. F7 remains PARTIAL; judge semantics and eligible interruption timing remain unavailable.

Further evidence: numbered documents 2 (product report) and 4 (GUI guide), plus Benchmarks/BENCHMARK_RESULTS.md. Editable Mermaid, PNG and SVG are in REFERENCES. The sanitized deployment copy has disclosed differences from the private measured installation.